

# Manual Deburring

## 1. Timesavers 10 Series Manual Grinder

**Manufacturer:** Timesavers

**Country / Region:** Netherlands / Europe

**Machine type:** Manual swing-arm deburring, edge-rounding, finishing machine

**Typical use:** Sheet-metal parts from laser cutting, punching, plasma cutting, waterjet cutting, and small-batch metal finishing.

### What this machine is for

The **Timesavers 10 Series** is an entry-level manual deburring machine. It is used when the operator wants more productivity and consistency than hand grinding, but does not need a full automatic through-feed deburring line. It is good for **flat metal parts**, especially small to medium sheet-metal parts.

The operator places the part on the worktable, fixes it using the high-friction table and vacuum area, then moves the grinding/brush head over the part manually. The rotating 180° head allows quick switching between consumables, such as a grinding disc and brush. Timesavers describes it as a cost-saving alternative for manual deburring, edge rounding, and finishing.

### Main applications

| Application         | Description                                                    |
|---------------------|----------------------------------------------------------------|
| Deburring           | Removes sharp burrs after cutting or punching                  |
| Edge rounding       | Softens sharp edges for safer handling and coating preparation |
| Laser oxide removal | Removes oxide layer after laser cutting                        |
| Slag removal        | Removes heavier cutting residue                                |
| Surface finishing   | Improves visual and functional surface quality                 |

### Key technical specifications

| Item                   | Specification                                                                                                               |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Model / configuration  | MG 1300 × 880                                                                                                               |
| Number of heads        | 2                                                                                                                           |
| Head types             | Disc + brush                                                                                                                |
| Disc / brush diameter  | 150 mm                                                                                                                      |
| Rotating grinding head | 180°                                                                                                                        |
| Worktable size         | 1300 × 800 mm                                                                                                               |
| Working height         | 960 mm                                                                                                                      |
| Vacuum area            | 260 × 310 mm                                                                                                                |
| Max part thickness     | 100 mm                                                                                                                      |
| Main motor             | 0.75 kW                                                                                                                     |
| Total power / supply   | 400 V / 2.5 kW / 11.6 A                                                                                                     |
| Machine weight         | Approx. 270 kg                                                                                                              |
| Standard features      | Movable arm with weight compensation, frequency-controlled drive, vacuum table, dust collection container, CE certification |
| Catalog / brochure     | Official Timesavers company brochure PDF available                                                                          |

Sources: Timesavers brochure confirms the 10 Series purpose, two-head setup, rotating shaft, vacuum table, table size, weight, motor, supply, max thickness, and applications.

## 2. NS Máquinas CL75 / CLF75

**Manufacturer:** NS Máquinas

**Country / Region:** Portugal / Europe

**Machine type:** Abrasive belt grinding / finishing machine

**Typical use:** Metal finishing, weld seam polishing, deburring, surface cleaning, and large sheet grinding.

Important note: **CL75 and CLF75 are different machine styles.**

The **CL75** is a stationary multipurpose belt grinder.

The **CLF75** is a mobile floor-use belt grinding machine for large metal sheets that are difficult to move.

---

## 2A. NS Máquinas CL75

### What this machine is for

The **CL75** is a compact abrasive belt machine for grinding, deburring, weld seam polishing, and finishing metal components. It is more like a **belt grinding workstation** than a flat-table swing-arm deburring machine. It can be used with contact roller grinding or top belt flat-section deburring. NS Máquinas also says centerless and notching modules can be connected to it.

### Main applications

| Application               | Description                                                |
|---------------------------|------------------------------------------------------------|
| Deburring                 | Removes burrs from metal parts using abrasive belt contact |
| Weld seam polishing       | Smooths welded joints                                      |
| Surface finishing         | Creates cleaner surface finish on metal parts              |
| Tube / component grinding | Can be used with contact roller setup                      |
| Optional expansion        | Centerless and notching modules available                  |

### Key technical specifications

| Item                | Specification        |
|---------------------|----------------------|
| Model               | CL75                 |
| Abrasive belt motor | 3 kW                 |
| Abrasive belt size  | 75 × 2000 mm         |
| Machine dimensions  | 1150 × 650 × 1100 mm |

|                    |                                                          |
|--------------------|----------------------------------------------------------|
| Weight             | 80 kg                                                    |
| Standard functions | Contact roller grinding, top belt flat-section deburring |
| Optional modules   | Centerless module, notching module                       |
| Catalog / brochure | NS Máquinas product page + catalog link available        |

Source: NS Máquinas lists CL75 purpose and specifications on its product page.

## 2B. NS Máquinas CLF75

What this machine is for

The **CLF75** is designed for **large metal sheets on the floor**. Instead of moving a heavy sheet to a machine, the operator moves the machine over or along the sheet. This makes it useful for large plates, heavy steel sheets, wide welded panels, and floor-level finishing jobs. NS Máquinas says it is used for cleaning metal surfaces, grinding calamine, belt finishing, polishing, and grinding weld seams.

Main applications

| Application          | Description                                        |
|----------------------|----------------------------------------------------|
| Large sheet grinding | For plates that are too large/heavy to move easily |
| Weld seam grinding   | Uses contact roller to grind weld seams            |
| Surface cleaning     | Removes calamine / scale from metal surface        |
| Belt finishing       | Creates linear abrasive finish                     |
| Mobile finishing     | Machine moves along the workpiece                  |

Key technical specifications

| Item                       | Specification                                               |
|----------------------------|-------------------------------------------------------------|
| Model                      | CLF75                                                       |
| Abrasive belt size         | 75 × 2000 mm                                                |
| Abrasive belt speed        | 30 m/s                                                      |
| Main motor                 | 4 kW                                                        |
| Total power                | 4 kW                                                        |
| Current consumption        | 9 A                                                         |
| Dust extraction connection | Ø80 mm per head                                             |
| Machine dimensions         | 1520 × 700 × 1030 mm                                        |
| Weight                     | 97 kg                                                       |
| Catalog / brochure         | NS Máquinas product page /<br>technical file link available |

Source: NS Máquinas lists CLF75 applications and technical details on its product page.

---

### 3. Knoppo KD-BB1308D

**Manufacturer:** Knoppo / Knoppo Automation

**Country / Region:** China

**Machine type:** Manual table-type deburring / sheet-metal grinding machine

**Typical use:** Manual deburring and edge finishing of flat sheet-metal parts.

What this machine is for

The **Knoppo KD-BB1308D** is a manual deburring machine for flat sheet-metal parts. It uses a disc/cup-type grinding head on a workbench with a vacuum holding area. The operator moves the grinding head manually over the part to remove burrs and soften edges.

This machine is similar in concept to the Timesavers 10 Series, but appears to be positioned as a lower-cost Chinese manual deburring option. The public Made-in-China listing provides technical parameters, but I did not find a full official manufacturer PDF brochure in the search results.

**Main applications**

| Application               | Description                                               |
|---------------------------|-----------------------------------------------------------|
| Sheet-metal deburring     | Removes burrs from laser-cut, punched, or cut sheet parts |
| Edge softening            | Reduces sharpness on part edges                           |
| Manual grinding           | Allows operator-controlled surface correction             |
| Small / medium flat parts | Best suited for flat workpieces on the table              |

#### Key technical specifications

| Item                        | Specification                                                                |
|-----------------------------|------------------------------------------------------------------------------|
| Model                       | KD-BB1308D                                                                   |
| Number of grinding heads    | 1                                                                            |
| Grinding head type          | Disc type / cup shape                                                        |
| Workbench size              | 1300 × 800 mm                                                                |
| Vacuum area                 | 300 × 250 mm                                                                 |
| Abrasive drive motor        | 230 V / 0.75 kW / 4 A                                                        |
| Grinding plate diameter     | 150 mm                                                                       |
| Grinding head rotation      | 180°                                                                         |
| Maximum workpiece thickness | 60 mm                                                                        |
| Catalog / brochure          | Public Made-in-China product page found; official PDF brochure not confirmed |

Source: Knoppo listing gives the KD-BB1308D main technical parameters including head count, workbench, vacuum area, motor, grinding plate diameter, head rotation, and max thickness.

---

## 4. RSM / eRSM RMD-150 and RMD-165

**Manufacturer:** RSM Machinery / eRSM

**Country / Region:** China

**Machine type:** Manual swing-arm deburring, edge-rounding, polishing machine

**Typical use:** Punched and laser-cut sheet-metal parts, deburring, edge rounding, polishing, oxide removal, slag removal.

---

### 4A. RSM RMD-150

**What this machine is for**

The **RMD-150** is the smaller manual swing-arm deburring model. It is useful for low-volume or specialized parts where automatic programming is not worth it. The operator manually guides the grinding/brush head over the part. It has adjustable grinding head speed, arm height/angle adjustment, and quick-change grinding heads.

**Main applications**

| Application           | Description                                               |
|-----------------------|-----------------------------------------------------------|
| Deburring             | Removes sharp burrs from flat parts                       |
| Simple chamfering     | Light edge beveling                                       |
| Grinding              | Surface correction and burr removal                       |
| Finishing             | Basic surface finishing                                   |
| Small curved surfaces | RSM says it can handle flat parts or tiny curved surfaces |

**Key technical specifications**

| Item                     | Specification  |
|--------------------------|----------------|
| Model                    | RMD-150        |
| Processing type          | Manual         |
| Material thickness       | 1–150 mm       |
| Cantilever working range | 1200 mm × 360° |

|                        |                                                                              |
|------------------------|------------------------------------------------------------------------------|
| Vacuum operating area  | 325 × 225 mm                                                                 |
| Sanding disc size      | Ø150 mm                                                                      |
| Brush size             | Ø150 mm                                                                      |
| Minimum material size  | ≥ L50 × W50 mm                                                               |
| Feed speed             | 0.5–6 m/min                                                                  |
| Weight                 | 170 kg                                                                       |
| Grinding head quantity | 1, based on RSM comparison table                                             |
| Grinding head power    | 1.2 kW, based on RSM comparison table                                        |
| Catalog / brochure     | RSM product page found; PDF download may be available from manufacturer page |

Sources: RSM RMD-150 page gives material thickness, cantilever range, vacuum area, disc/brush size, minimum part size, feed speed, and weight. RSM's RMD comparison table lists RMD-150 as one grinding head with 1.2 kW head power.

---

## 4B. RSM RMD-165

What this machine is for

The **RMD-165** is the larger / higher-capability manual deburring model. It has a larger cantilever range and dual grinding heads, so one head can be used for coarse deburring and another for finer finishing or edge rounding. It is a practical middle-ground machine between hand tools and a more expensive automatic deburring line.

RSM describes it for deburring, edge rounding, and polishing of punched and laser-cut parts. It uses a pneumatic-floating grinding head and vacuum-chuck worktable to make manual operation easier.

Main applications

| Application | Description |
|-------------|-------------|
|-------------|-------------|



|                           |                                                         |
|---------------------------|---------------------------------------------------------|
| Deburring                 | Removes burrs from laser-cut / punched parts            |
| Edge rounding             | Creates safer, smoother edges                           |
| Polishing                 | Improves surface appearance                             |
| Oxide removal             | Removes oxide layer from laser-cut edges                |
| Slag removal              | Removes heavier cutting residue                         |
| Oversized part processing | Swing arm can process parts larger than table footprint |

#### Key technical specifications

| Item                          | Specification         |
|-------------------------------|-----------------------|
| Model                         | RMD-165               |
| Processing type               | Manual                |
| Material thickness            | 1–150 mm              |
| Cantilever working range      | 1500 mm × 360°        |
| Vacuum operating area         | 330 × 300 mm          |
| Sanding disc size             | Ø150 mm               |
| Brush size                    | Ø165 mm               |
| Grinding head quantity        | 2                     |
| Grinding head power           | 0.75 kW               |
| Machine size                  | 1700 × 1700 × 1700 mm |
| Upper platform workpiece size | 1000 × 800 mm         |
| No-platform workpiece range   | 1500 × 1500 mm        |

|                    |                                                                                                                                   |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Key features       | Pneumatic-floating grinding head, vacuum-chuck table, quick-change consumables, dual heads, swing-arm design, touchscreen control |
| Catalog / brochure | RSM page has “Download Datasheet,” but the direct PDF fetch was not accessible through my browser tool                            |

Sources: RSM product page provides RMD-165 thickness, working range, vacuum area, disc/brush size, head quantity, size, power, and comparison table. A distributor page also lists RMD-165 specs including 330 × 300 mm vacuum area, 1000 × 800 mm table workpiece size, 1500 × 1500 mm swing-arm range, 1–150 mm thickness, 0.75 kW head power, 2 heads, and 650 kg packing weight.

## Comparison Summary

| Machine              | Origin               | Best for                                        | Head / Tool Type      | Work Area / Table        | Thickness              | Power           | Notes                                        |
|----------------------|----------------------|-------------------------------------------------|-----------------------|--------------------------|------------------------|-----------------|----------------------------------------------|
| Timesavers 10 Series | Europe / Netherlands | Professional manual deburring and edge rounding | 2 heads: disc + brush | 1300 × 800 mm table      | Up to 100 mm           | 2.5 kW total    | Strongest official documentation among these |
| NS CL75              | Europe / Portugal    | Belt grinding, weld seam polishing, composite   | Abrasive belt         | Not a table-type machine | Not publicly confirmed | 3 kW belt motor | Multipurpose belt grinder                    |

|                               |                             |                                                                    |                                       |                                        |                                  |                    |                                                                                   |
|-------------------------------|-----------------------------|--------------------------------------------------------------------|---------------------------------------|----------------------------------------|----------------------------------|--------------------|-----------------------------------------------------------------------------------|
|                               |                             | nent<br>deburri<br>ng                                              |                                       |                                        |                                  |                    |                                                                                   |
| NS<br>CLF75                   | Europe<br>/<br>Portug<br>al | Large<br>sheet<br>grindin<br>g on<br>floor                         | Abrasiv<br>e belt                     | Mobile<br>floor-<br>use<br>machin<br>e | Not<br>publicly<br>confirm<br>ed | 4 kW               | Good<br>for<br>heavy/l<br>arge<br>sheets                                          |
| Knopp<br>o KD-<br>BB1308<br>D | China                       | Budget<br>manual<br>table<br>deburri<br>ng                         | 1<br>disc/cu<br>p head                | 1300 ×<br>800<br>mm<br>table           | Up to<br>60 mm                   | 0.75<br>kW         | Similar<br>concep<br>t to<br>Timesa<br>vers,<br>fewer<br>verified<br>feature<br>s |
| RSM<br>RMD-<br>150            | China                       | Small/<br>manual<br>deburri<br>ng<br>workst<br>ation               | 1 head,<br>Ø150<br>tools              | 1200<br>mm<br>swing<br>range           | 1–150<br>mm                      | 1.2 kW<br>head     | Compa<br>ct<br>model                                                              |
| RSM<br>RMD-<br>165            | China                       | Larger<br>manual<br>deburri<br>ng with<br>coarse/<br>fine<br>heads | 2<br>heads,<br>Ø150/<br>Ø165<br>tools | 1500<br>mm<br>swing<br>range           | 1–150<br>mm                      | 0.75<br>kW<br>head | Better<br>product<br>ivity<br>than<br>RMD-<br>150                                 |

### Simple recommendation

For **best verified European manual deburring machine**, choose **Timesavers 10 Series**. It has a real brochure, clear specs, CE certification, vacuum table, 2 heads, and professional sheet-

metal deburring focus.

For **belt grinding and weld seam finishing**, choose **NS CL75 or CLF75**. They are not the same style as Timesavers/RSM/Knoppo. CL75 is more of a stationary belt grinder; CLF75 is for large sheets on the floor.

For **lower-cost Chinese manual table deburring**, choose **Knoppo KD-BB1308D** or **RSM RMD series**. Between RSM models, **RMD-165** is more capable than RMD-150 because it has a larger swing range and dual heads.

okay, give me the download link for each machine

I'll look for direct manufacturer brochure/datasheet downloads first, then use reliable product pages only where a PDF download is not publicly reachable.

Stopped thinking

okay, give me the download link for each machine catalog

Thought for 22s

Here are the **catalog / datasheet download links** I found.

| Machine                     | Catalog / Datasheet link                          | Status                                                                                        |
|-----------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Timesavers 10 Series</b> | <a href="#">Download Timesavers brochure PDF</a>  | Official PDF brochure. The 10 Series is included in this brochure.                            |
| <b>NS Máquinas CL75</b>     | Download CL75 Technical File PDF                  | Official NS Máquinas document link shown on the CL75 page as "Bases Catalogue / English PDF." |
| <b>NS Máquinas CLF75</b>    | <a href="#">Download CLF75 Technical File PDF</a> | Official PDF technical file.                                                                  |
| <b>Knoppo KD-BB1308D</b>    | <a href="#">Official Knoppo product page</a>      | I found the official product page with technical parameters, but I did <b>not</b> find a      |

|                                   |                                               |                                                                                                                                |
|-----------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
|                                   |                                               | public PDF catalog download.                                                                                                   |
| <b>RSM / eRSM RMD-150</b>         | <a href="#">RMD-150 official product page</a> | Page has a “Download Datasheet” button, but the PDF fetch was unstable from my side. Specs are visible on the page.            |
| <b>RSM / eRSM RMD-165</b>         | <a href="#">RMD-165 official product page</a> | Official product page. Also covered in the RMD manual deburring comparison page.                                               |
| <b>RSM RMD Series PDF catalog</b> | <a href="#">Download RSM catalog PDF</a>      | This is the PDF behind RSM’s “Download Datasheet” button. My browser tool detected the link but could not fully fetch the PDF. |

For **Knoppo KD-BB1308D**, the safest usable “catalog” link is the official product page, not a PDF.

give me the recent prices for each machine

I’ll check current public price signals and separate **published prices** from **quote-only / estimated market ranges**. For industrial machines, many manufacturers hide prices, so I’ll be careful not to invent exact numbers.

Thought for 19s

Below are the **recent public price signals I found**. For these industrial machines, many official manufacturer pages are “ask for quote”, so treat the numbers as **FOB / marketplace reference prices**, not final delivered prices.

| Machine | Recent public price / range | Confidence | Notes |
|---------|-----------------------------|------------|-------|
|---------|-----------------------------|------------|-------|

|                                                           |                                                         |        |                                                                                                                                                                                                     |
|-----------------------------------------------------------|---------------------------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Timesavers 10 Series</b>                               | <b>Quote only</b> from official/dealer pages            | Medium | Official/dealer pages show “contact us / price info,” not a public price. Truman Systems lists the machine but no price; used-machine listings also show “price info” only.                         |
| <b>Timesavers-style Chinese equivalent / ADV 508-M300</b> | <b>US\$5,400 / set, or US\$4,500 / set for 10+ sets</b> | Medium | This is <b>not confirmed as genuine Timesavers</b> . It is an Alibaba listing by Anhui Advanced Grinding Tools, model <b>ADV 508-M300</b> , using a very similar swing-arm manual deburring format. |
| <b>NS Máquinas CL75</b>                                   | <b>Quote only</b>                                       | Medium | NS Máquinas/Base product pages use “Ask for a quote,” and I did not find a reliable current public price.                                                                                           |

|                           |                                                            |                            |                                                                                                                                                                                                                                        |
|---------------------------|------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NS Máquinas CLF75</b>  | <b>New: quote only; used 2014 listing: about US\$3,334</b> | Medium-low                 | New CLF75 supplier pages do not show a public price. One used listing shows a <b>2014 NS Máquinas CLF75</b> at <b>\$3,334</b> , but it is a used-equipment listing and may be outdated or unavailable.                                 |
| <b>Knoppo KD-BB1308D</b>  | <b>Quote only</b> from official Knoppo page                | Medium                     | Knoppo's official page says "Get the best price / quote now," but does not publish a price. Similar Chinese manual deburring machines commonly appear around <b>US\$4,500–6,000</b> , but I did not find a confirmed KD-BB1308D price. |
| <b>RSM / eRSM RMD-150</b> | <b>FOB US\$3,450–3,500 / piece</b>                         | High for marketplace price | RSM's Made-in-China listings show <b>RMD-150</b> at <b>US\$3,450 / piece</b> , and similar RMD-                                                                                                                                        |

|                           |                                                                       |                            |                                                                                                                                      |
|---------------------------|-----------------------------------------------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
|                           |                                                                       |                            | 150/manual deburring listings around <b>US\$3,500 / piece.</b>                                                                       |
| <b>RSM / eRSM RMD-165</b> | <b>US\$6,000 / set, US\$5,900 for 2–4 sets, US\$5,600 for 5+ sets</b> | High for marketplace price | Alibaba listing from RSM Machinery shows this pricing for <b>RMD-165</b> . Shipping fee and delivery date are negotiated separately. |

### Practical buying estimate

For budgeting, I would use these rough ranges:

| Machine              | Budget estimate before shipping/import/tax                   |
|----------------------|--------------------------------------------------------------|
| Timesavers 10 Series | <b>Ask dealer; likely higher than Chinese models</b>         |
| NS CL75              | <b>Ask NS dealer; likely quote-based</b>                     |
| NS CLF75             | <b>Ask NS dealer; used units may appear around US\$3k–5k</b> |
| Knoppo KD-BB1308D    | <b>Estimate US\$4,500–6,000</b>                              |
| RSM RMD-150          | <b>US\$3,450–3,500 FOB</b>                                   |
| RSM RMD-165          | <b>US\$5,600–6,000 FOB</b>                                   |

**Important:** FOB price usually excludes sea/air freight, import duty, customs broker fee, local delivery, installation, dust collector, spare abrasive tools, and training. Final landed price can



easily be **20–60% higher** depending on country and shipping method.

## How they work

You’re right — I gave specs and prices, but not enough explanation of **how each machine actually works**. Here is the user-friendly working principle for each one.

---

### 1. Timesavers 10 Series Manual Grinder

#### How it works

The **Timesavers 10 Series** is a **manual swing-arm deburring machine**.

The operator places a flat metal part on the machine table. The table has a **vacuum holding area**, so the part can be fixed in place during grinding. Then the operator moves the arm by hand over the part.

At the end of the arm, there is a rotating tool head. The head usually uses a **grinding disc** and a **brush**.

#### Simple workflow

| Step | What happens                                               |
|------|------------------------------------------------------------|
| 1    | Operator places the metal part on the table                |
| 2    | Vacuum holds the part so it does not move                  |
| 3    | Operator lowers the grinding head onto the surface or edge |
| 4    | The grinding disc removes burrs or sharp material          |
| 5    | The brush rounds or smooths the edge                       |
| 6    | Dust and metal particles are collected by the dust system  |
| 7    | Operator checks the part and repeats if needed             |

## What the tools do

| Tool          | Function                                                    |
|---------------|-------------------------------------------------------------|
| Grinding disc | Removes stronger burrs, sharp points, and rough edges       |
| Brush         | Rounds edges and makes the part safer to touch              |
| Vacuum table  | Holds the part during manual operation                      |
| Swing arm     | Lets the operator move the head smoothly over the workpiece |

## Good for

Timesavers 10 Series is good for:

- laser-cut sheet metal
- punched parts
- plasma-cut parts
- waterjet-cut parts
- small and medium flat parts
- parts that need safe rounded edges before painting or coating

## Not ideal for

It is not the best choice for:

- very high-volume automatic production
- complex 3D parts
- deep internal holes that the brush cannot reach well
- very heavy slag removal compared with heavy-duty slag grinders

---

## 2. NS Máquinas CL75

### How it works

The **NS CL75** is not a flat-table swing-arm deburring machine like the Timesavers 10 Series. It is more like a **belt grinding and finishing workstation**.

It uses a long abrasive belt that rotates continuously at high speed. The operator brings the metal part to the belt and grinds the surface, weld seam, edge, or corner.

#### Simple workflow

| Step | What happens                                                    |
|------|-----------------------------------------------------------------|
| 1    | Operator installs the correct abrasive belt                     |
| 2    | Machine rotates the belt continuously                           |
| 3    | Operator holds the workpiece against the belt                   |
| 4    | The abrasive belt removes burrs, weld marks, or surface defects |
| 5    | Operator changes the part angle depending on the area to grind  |
| 6    | Finished part is visually checked                               |

#### What the belt does

The abrasive belt acts like a moving sandpaper surface.

| Belt area           | Function                                                            |
|---------------------|---------------------------------------------------------------------|
| Contact roller area | Stronger grinding pressure, useful for weld seams and edges         |
| Flat belt area      | Smoother finishing and light deburring                              |
| Optional modules    | Can support tube notching or centerless grinding depending on setup |

#### Good for

NS CL75 is good for:

- grinding small metal parts
- deburring edges manually
- removing weld seams
- polishing welded joints
- finishing tubes, profiles, and flat components

- workshop repair and fabrication jobs

### Not ideal for

It is not ideal for:

- large flat sheet deburring in batch production
- automatic edge rounding on many identical parts
- holding very small parts safely by hand without fixtures
- parts that need highly repeatable uniform edge radius

### Main difference from Timesavers

| Timesavers 10 Series        | NS CL75                               |
|-----------------------------|---------------------------------------|
| Workpiece stays on table    | Operator holds part against belt      |
| Swing arm moves over part   | Abrasive belt stays in one place      |
| Better for flat sheet parts | Better for general workshop grinding  |
| Has vacuum holding table    | Usually hand-held workpiece operation |

## 3. NS Máquinas CLF75

### How it works

The **NS CLF75** is a **mobile belt grinding machine**.

It is designed for large metal sheets or plates that are too heavy to lift onto a normal machine. Instead of moving the metal sheet to the grinder, the operator moves the grinder over the sheet.

Think of it like a heavy-duty belt grinder on a trolley.

### Simple workflow

| Step | What happens                                             |
|------|----------------------------------------------------------|
| 1    | Large metal plate is placed on the floor or work support |
| 2    | Operator positions the CLF75 near the grinding area      |

|   |                                                            |
|---|------------------------------------------------------------|
| 3 | Abrasive belt starts rotating                              |
| 4 | Operator moves the machine across the surface or weld seam |
| 5 | Belt removes scale, weld seam, burrs, or surface defects   |
| 6 | Dust extraction removes grinding dust                      |
| 7 | Operator repeats passes until finish is acceptable         |

### What it does

| Function               | Explanation                                          |
|------------------------|------------------------------------------------------|
| Surface cleaning       | Removes oxide, calamine, and scale from metal plates |
| Weld seam grinding     | Smooths welded areas on large panels                 |
| Surface finishing      | Creates a brushed or ground finish                   |
| Large sheet processing | Works on parts too large to move easily              |

### Good for

NS CLF75 is good for:

- large steel plates
- heavy metal sheets
- welded tanks or panels
- large fabrication parts
- plate surface cleaning
- weld seam finishing on big parts

### Not ideal for

It is not ideal for:

- small parts
- precision edge rounding

- automatic deburring
- small laser-cut components
- internal holes and small contours

#### Main difference from CL75

| CL75                        | CLF75                                |
|-----------------------------|--------------------------------------|
| Stationary machine          | Mobile trolley-type machine          |
| Operator moves part to belt | Operator moves machine over part     |
| Good for small/medium parts | Good for large plates                |
| Workshop grinder            | Large sheet / floor grinding machine |

## 4. Knoppo KD-BB1308D

### How it works

The **Knoppo KD-BB1308D** is also a **manual swing-arm deburring machine**, similar in concept to the Timesavers 10 Series.

The part is placed on the worktable. A vacuum area helps hold the workpiece. The operator manually moves the grinding head over the sheet-metal part.

The machine uses a **disc or cup-shaped grinding tool** to remove burrs and smooth the edge.

### Simple workflow

| Step | What happens                                      |
|------|---------------------------------------------------|
| 1    | Operator places the sheet-metal part on the table |
| 2    | Vacuum holds the part in place                    |
| 3    | Operator moves the swing arm over the part        |
| 4    | Rotating disc contacts the burr or sharp edge     |
| 5    | Burr is removed by abrasion                       |
| 6    | Operator repeats passes until edge is smooth      |

|   |                                                                  |
|---|------------------------------------------------------------------|
| 7 | Dust is removed by connected dust collector or collection system |
|---|------------------------------------------------------------------|

### What the grinding head does

The grinding head spins a disc/cup wheel. When the tool touches the part, it removes small amounts of metal from sharp edges and burrs.

| Part          | Function                             |
|---------------|--------------------------------------|
| Swing arm     | Allows manual movement over the part |
| Disc/cup head | Removes burrs and sharp edges        |
| Vacuum area   | Holds part during operation          |
| Worktable     | Supports flat sheet-metal parts      |

### Good for

Knoppo KD-BB1308D is good for:

- basic sheet-metal deburring
- small and medium flat parts
- laser-cut or punched parts
- lower-volume production
- companies wanting a lower-cost manual solution

### Not ideal for

It is not ideal for:

- very high throughput
- fully automatic production
- complex 3D parts
- very consistent edge-radius requirements
- thick slag removal

### Main difference from Timesavers

|                      |                   |
|----------------------|-------------------|
| Timesavers 10 Series | Knoppo KD-BB1308D |
|----------------------|-------------------|

|                             |                                        |
|-----------------------------|----------------------------------------|
| European professional brand | Chinese lower-cost option              |
| Usually 2-head setup        | Public info shows 1-head setup         |
| Better documented           | Less public official PDF documentation |
| Higher likely price         | Lower likely price                     |

## 5. RSM RMD-150

### How it works

The **RSM RMD-150** is a compact **manual swing-arm deburring machine**.

The operator places the workpiece on the vacuum table. The swing arm has one grinding/polishing head. The operator moves the arm manually over the edges and surface of the part.

It can use a sanding disc or brush depending on the job.

### Simple workflow

| Step | What happens                           |
|------|----------------------------------------|
| 1    | Operator places workpiece on table     |
| 2    | Vacuum table holds the part            |
| 3    | Operator selects sanding disc or brush |
| 4    | Operator moves the arm over the part   |
| 5    | Disc removes burrs or sharp edges      |
| 6    | Brush smooths or rounds the edge       |
| 7    | Operator checks the edge quality       |

### What it does

| Tool         | Function                           |
|--------------|------------------------------------|
| Sanding disc | Stronger burr removal and grinding |



|                  |                                 |
|------------------|---------------------------------|
| Brush            | Edge rounding and polishing     |
| Vacuum table     | Keeps small/flat parts stable   |
| Single swing arm | Manual movement across the part |

### Good for

RMD-150 is good for:

- small workshops
- low to medium-volume sheet-metal deburring
- small laser-cut parts
- punched parts
- basic chamfering
- simple edge rounding

### Not ideal for

It is not ideal for:

- very large sheets
- very high-volume production
- strong automation needs
- heavy slag removal
- fast switching between coarse and fine tools compared with dual-head machines

---

## 6. RSM RMD-165

### How it works

The **RSM RMD-165** works like the RMD-150, but it is larger and usually more capable.

The main advantage is that it has **two grinding heads**. This means the operator can use one head for rough deburring and another head for finishing or edge rounding.

For example:

- Head 1: sanding disc for burr removal
- Head 2: brush for edge rounding / polishing

This reduces tool-changing time and makes the process faster.

### Simple workflow

| Step | What happens                                      |
|------|---------------------------------------------------|
| 1    | Operator places the sheet-metal part on the table |
| 2    | Vacuum fixes the part                             |
| 3    | Operator uses first head for rough burr removal   |
| 4    | Operator rotates or changes to second head        |
| 5    | Second head rounds or polishes the edge           |
| 6    | Dust is extracted                                 |
| 7    | Finished part is checked                          |

### What the two heads do

| Head   | Typical function                          |
|--------|-------------------------------------------|
| Head 1 | Coarse grinding / burr removal            |
| Head 2 | Fine brushing / edge rounding / polishing |

### Good for

RMD-165 is good for:

- larger sheet-metal parts than RMD-150
- laser-cut parts
- punched parts
- deburring and edge rounding in one process
- shops that need better productivity but still want manual control
- small-batch and mixed-part production

### Not ideal for

It is not ideal for:

- fully automatic production lines
- extremely high-volume parts

- complex 3D parts
- very precise automatic edge-radius control
- parts requiring CNC-controlled consistency

#### Main difference between RMD-150 and RMD-165

| RMD-150                   | RMD-165                                   |
|---------------------------|-------------------------------------------|
| Smaller model             | Larger model                              |
| Usually 1 grinding head   | Usually 2 grinding heads                  |
| Lower cost                | Higher cost                               |
| Good for simple deburring | Better for deburring + polishing/rounding |
| Smaller working range     | Larger swing-arm range                    |

#### Overall working-method comparison

| Machine                     | How it works                                                   | Best use                                              |
|-----------------------------|----------------------------------------------------------------|-------------------------------------------------------|
| <b>Timesavers 10 Series</b> | Operator moves swing-arm head over vacuum-held sheet part      | Professional manual deburring / edge rounding         |
| <b>NS CL75</b>              | Operator pushes part against moving abrasive belt              | General grinding, weld polishing, component finishing |
| <b>NS CLF75</b>             | Operator moves mobile belt grinder over large sheet/plate      | Large plate grinding and weld seam finishing          |
| <b>Knoppo KD-BB1308D</b>    | Operator moves swing-arm disc head over vacuum-held sheet part | Budget manual sheet-metal deburring                   |
| <b>RSM RMD-150</b>          | Operator uses single-head swing arm over vacuum-held part      | Compact manual deburring                              |
| <b>RSM RMD-165</b>          | Operator uses dual-head swing arm for rough + finish           | Larger manual deburring with better productivity      |

## Easiest way to understand them

There are really **3 machine families** here:

### 1. Swing-arm table deburring machines

Includes:

- Timesavers 10 Series
- Knoppo KD-BB1308D
- RSM RMD-150
- RSM RMD-165

These are for **flat sheet-metal parts**.

The part stays on the table, and the operator moves the grinding/brush head over it.

### 2. Stationary belt grinding machine

Includes:

- NS CL75

This is for **general workshop grinding and finishing**.

The belt stays in place, and the operator brings the part to the belt.

### 3. Mobile belt grinding machine

Includes:

- NS CLF75

This is for **large plates or heavy sheets**.

The part stays on the floor/table, and the operator moves the whole machine over the part.